(a) if $\vec{h} = \vec{r} \times \vec{v} = \vec{r} \times \vec{r}'$ then finding $\vec{r} \times \vec{r}'$ should give $\vec{h}$ in terms of r and θ and $\frac{dr}{dt}$ and $\frac{d\theta}{dt}$ in the orthoginal direction to both $\vec{r}$ and $\vec{v}$ which is the $\hat{k}$ direction.

$$\vec{r} = \langle r\cos\theta, r\sin\theta \rangle = r\langle \cos\theta, \sin\theta \rangle$$

$$\vec{r}' = \left\langle \frac{dr}{dt}(r\cos\theta) + \frac{d\theta}{dt}(r\cos\theta), \frac{dr}{dt}(r\sin\theta) + \frac{d\theta}{dt}(r\sin\theta) \right\rangle$$

$$\vec{v} = \vec{r}' = \langle r'\cos\theta - \theta' r\sin\theta, r'\sin\theta + \theta' r\cos\theta \rangle$$

$$\frac{d\theta}{dt} = \theta'$$

$$\vec{r} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r\cos\theta & r\sin\theta & 0 \\ (r'\cos\theta - \theta' r\sin\theta) & (r'\sin\theta + \theta' r\cos\theta) & 0 \end{vmatrix} =$$

$$= 0\hat{i} - 0\hat{j} + \left(r\cos\theta(r'\sin\theta + \theta' r\cos\theta) - r\sin\theta(r'\cos\theta - \theta' r\sin\theta)\right)\hat{k}$$

distributing gets

$$\vec{r} \times \vec{v} = (\theta' r^2\cos\theta^2 + \theta' r^2\sin\theta^2 - (r r'\sin\theta\cos\theta + r r'\sin\theta\cos\theta))\hat{k} = (r^2\theta'(\cos\theta^2 + \sin\theta^2))\hat{k}$$

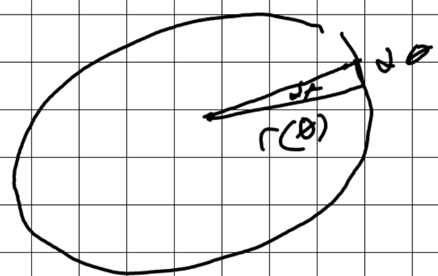
$$\vec{r} \times \vec{v} = r^2\theta'\hat{k} = r^2\frac{d\theta}{dt}\hat{k}$$

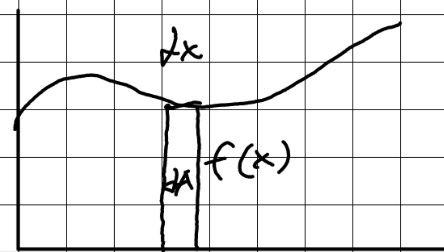
(b) If $|\vec{h}| = h$ then $\left|\left\langle 0, 0, r^2\frac{d\theta}{dt} \right\rangle\right| = h$

$$h = \sqrt{0^2 + 0^2 + \left(r^2\frac{d\theta}{dt}\right)^2} = r^2\frac{d\theta}{dt}$$

$$h = r^2\frac{d\theta}{dt}$$

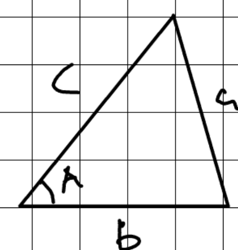
for the area of a polar curve you add up an infinite amount of triangles that are infinitly small much in the same way you would add up rectangles in the cartisian plane.





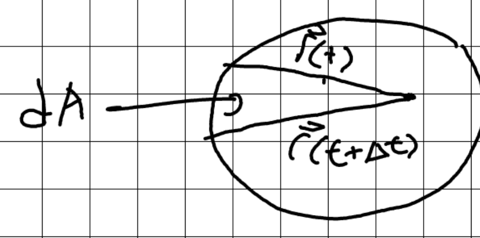
The Area of a triangle can be found with the formula:

$$\text{Area} = \frac{cb\sin(A)}{2} \rightarrow$$



In our case the length of the two sides are the magnitudes of the two vectors and the angle is the angle betweech them. This is equal to the magnitude of the cross product of the two vectors devided by two. In this case the triangle represents the change in the area, or dA.

$$dA = \frac{\left|\vec{r}(t) \times \vec{r}(t+\Delta t)\right|}{2}$$



The vector $\vec{r}(t+\Delta t)$ describes the position after a small change in time we can describe this vector as a sum of a Taylor sieries of the position function $\vec{r}(t)$ at point t.

$$\vec{r}(t+\Delta t)=\sum_{n=0}^{\infty}\frac{\vec{r}^{n}(t)}{n!}((t+\Delta t)-t)^{n}=\frac{\vec{r}(t)}{1}+\frac{\vec{v}(t)\Delta t}{1}+\frac{\vec{a}(t)\Delta t^{2}}{2}\dots$$

If this infinite kinimatics sum is plugged in for $\vec{r}(t+\Delta t)$ and distributed you get:

$$\frac{1}{2}\left|\vec{r}(t)\times\vec{r}(t+\Delta t)\right|=\left|\vec{r}(t)\times\left(r(t)+\vec{v}(t)\Delta t+\vec{a}(t)\frac{\Delta t^{2}}{2}\dots\right)\right|=$$

$$\frac{1}{2}\left|\vec{r}(t)\times\vec{r}(t)+\vec{r}(t)\times\vec{v}(t)\Delta t+\vec{r}(t)\times\vec{a}(t)\frac{(\Delta t)^{2}}{2}\dots\right|=dA$$

To find $\frac{dA}{dt}$ we most devide the area of the triangle, dA, by the change in time and take the limit as time aproaches zero.

$$\frac{dA}{dt}=\lim_{\Delta t\to 0}\left(\frac{\frac{1}{2}\left|\vec{r}(t)\times\vec{r}(t)+\vec{r}(t)\times\vec{v}(t)\Delta t+\vec{r}(t)\times\vec{a}(t)\frac{(\Delta t)^{2}}{2}\dots\right|}{\Delta t}\right)$$

With the fact that $(\vec{a}b)\times(\vec{c}d)=(b\cdot d)(\vec{a}\times\vec{c})$ The Δt term can be factored out of each of these cross products.

$$\frac{dA}{dt}=\lim_{\Delta t\to 0}\left(\frac{\frac{1}{2}\left|(\vec{r}(t)\times\vec{r}(t))+\Delta t\,(\vec{r}(t)\times\vec{v}(t))+\frac{\Delta t^{2}}{2}(\vec{r}(t)\times\vec{a}(t))\dots\right|}{\Delta t}\right)$$

this Δt can then be distributed into the magnitude and because $\vec{r}(t)\times\vec{r}(t)=0$ it can be removed to get this limit.

$$\frac{dA}{dt}=\lim_{\Delta t\to 0}\left(\frac{1}{2}\left|\vec{r}(t)\times\vec{v}(t)+\frac{(\vec{r}(t)\times\vec{a}(t))\Delta t}{2}+\frac{(\vec{r}(t)\times\vec{a}'(t))\Delta t^{2}}{6}\dots\right|\right)$$

Now Δt can be replaced with 0 to get the equation:

$$\frac{dA}{dt} = \frac{1}{2}\left|\vec{r}(t) \times \vec{v}(t)\right|$$

because we have found that $h = \left|\vec{r}(t) \times \vec{v}(t)\right|$ and $h = r^2 \frac{d\theta}{dt}$:

$$\frac{dA}{dt} = \frac{1}{2}\left|\vec{r}(t) \times \vec{v}(t)\right| = \frac{h}{2} = \frac{r^2}{2}\frac{d\theta}{dt}$$

Since h has been found to be constant and $\frac{dA}{dt} = h \cdot \frac{1}{2}$ it must be the case that $\frac{dA}{dt}$ is a constant, proving keplers 2nd Law.

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2. The formula for the area of an ellipse is $ab\pi$ where a and b are the semi-major and semi-minor axis respectivly. Because we have shown that the relationship between the time traveled and the total area is linear (h is costant), to find the time it takes to travel around the ellipse we can simply divivide the total area by the constant rate of change $\left(\frac{h}{2}\right)$:

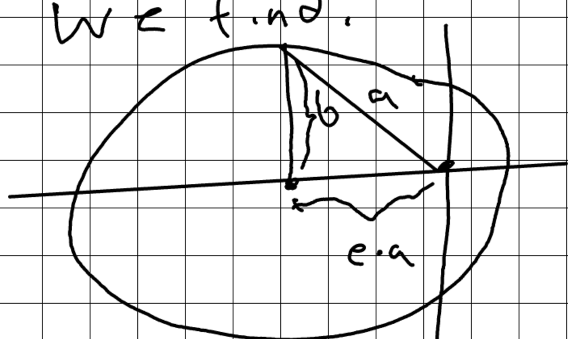
$$T = \frac{ab\pi}{\left(\frac{h}{2}\right)} = \frac{2\pi ab}{h}$$

$\vec{r}(\theta) = \frac{h^2/GM}{1+e\cos\theta}$ is the equation for the the location vector derivede from newtons Laws.

$r(\theta) = \frac{a(1-e^2)}{1+e\cos(\theta)}$ is the equation for of any ellipse where the foci is the origin where a is the semi-major axis. From those we can solve for a

$$\frac{h^2/GM}{1+e\cos\theta} = \frac{a(1-e^2)}{1+e\cos\theta} \rightarrow \frac{h^2}{GM} = a(1-e^2)$$

Since the distance from the center to the foci of an ellipse is e·a and the distance from the foci to the top of the ellipse is using a triangle we find:



$$b^2 = a^2 - (ea)^2$$

$$\downarrow$$

$$b^2 = a^2(1-e^2)$$

$$\downarrow$$

$$b^2/a = a(1-e^2)$$

Thus, $\frac{b^2}{a} = \frac{h^2}{GM}$

The formula for a general Conic Polar n equation is the following:

$$r = \frac{ed}{1+e\cos\theta}$$

by equating this to the $\vec{r}(\theta)$ equation from newtons laws we get:

$$\frac{ed}{1+e\cos\theta} = \frac{h^2/GM}{1+e\cos\theta} \rightarrow ed = \frac{h^2}{GM}$$

Thus

$$ed = \frac{h^2}{GM} = \frac{b^2}{a}$$

From that we find that:

$$T^2 = \frac{4\pi^2 a^2 b^2}{h^2} = \frac{4\pi^2 a^2 b^2}{\left(GM\left(\frac{b^2}{a}\right)\right)} = \frac{4\pi^2}{GM}a^3$$

3

$$T^2 = \frac{4\pi^2}{GM} a^3 \rightarrow \frac{GMT^2}{4\pi^2} = a^3$$

$$\rightarrow a = \sqrt[3]{\frac{GMT^2}{4\pi^2}} \rightarrow 2a = 2\sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

Solving for 2a lets us find the major axis by plugging in our known values:

$$2a = 2\sqrt[3]{\frac{(6.67\cdot 10^{-11})(1.99\cdot 10^{30})(365.25\cdot 60^2\cdot 24)^2}{4\pi^2}}$$

$2a \approx 2.99\times 10^{11} \approx$ major axis of earth

4

$$a = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

The a in this equation represents the length from the center of mass to the farthest point the satilite reaches given a Period, Mass, and the gravitational constant G. For the object to stay above a specific point it must have a period of 1 day and be equal distances from the earth at all points. This means a minus the radius of the earth will get us the altitude at all points.

$$Altitude = \sqrt[3]{\frac{GMT^2}{4\pi^2}} - R_{earth}$$

$$Altitude = \sqrt[3]{\frac{(6.67 \cdot 10^{-11})(5.98 \cdot 10^{24})(24 \cdot 60^2)^2}{4\pi^2}} - 6.37 \cdot 10^6$$

$$Altitude \approx 35880.474 \text{ Km}$$